Physics-Guided Invariant Learning for LPBF Defect Segmentation

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Abstract

Three Scientific Machine Learning mechanisms are applied to LPBF powder bed defect segmentation, each embedding the governing Volumetric Energy Density equation $\mathrm{VED} = P/(v \cdot h \cdot t)$ in a different way. NOTEARS causal structure learning recovers the process graph $\{P, v, h, t\} \to \mathrm{VED} \to$ defect from data. Invariant Risk Minimisation (IRM) stratifies UNet training by VED regime, yielding interpretable failure modes rather than a single aggregate metric. A Physics-Informed Neural Network (PINN) encodes the defect-nucleation ODE as a residual loss term, and a physics-constrained Neural ODE models layer-wise defect evolution. On the PB dataset (2,638 images, EOS M290, 316L SS), the baseline UNet achieves Dice = 0.787 and ROC AUC = 0.9955. The PINN ablation shows that a physics penalty weight of $\lambda = 1$ reduces MSE against the ODE ground truth relative to a data-only baseline.

1 Introduction

In Laser Powder Bed Fusion, the dominant process variable is Volumetric Energy Density:

$$\mathrm{VED} = \frac{P}{v \cdot h \cdot t}, \tag{1}$$

where P is laser power, v scan speed, h hatch spacing, and t layer thickness. Below a critical VED threshold ($\approx 30\,\mathrm{J/mm^3}$ for 316L SS), powder fails to fully melt and lack-of-fusion (LoF) pores form [1].

Standard segmentation pipelines treat defect detection as a purely data-driven problem. The physics is never consulted. The question here is whether building Eq. (1) into the learning pipeline changes anything meaningful, and if so, how.

Three SciML mechanisms are tested. NOTEARS [3] is used to recover the causal graph over process variables from data, without being told the VED formula. IRM [4] is used to train a UNet with a penalty that forces invariance across VED-defined environments. A PINN [6] encodes the nucleation ODE as a soft constraint evaluated at collocation points, requiring no additional image labels. Each mechanism embeds the physics differently, and the payoffs are compared.

2 Method

2.1 UNet Baseline

A standard four-level UNet [2] is used, with channel depths $64 \to 128 \to 256 \to 512$ and a 1024-channel bottleneck. Ten one-factor-at-a-time experiments vary learning rate (10^{-4}, 5×10^{-4}, 10^{-3}), loss function (BCE, Dice, BCEDice, Focal [5]), optimiser (Adam, SGD, RMSprop), and threshold $\tau \in \{0.3, 0.5, 0.7\}$. Batch size is 4, with 50 epochs maximum and patience 7. The combined BCEDice loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCE}} + 1 - \frac{2\sum_i y_i \hat{p}_i + \varepsilon}{\sum_i y_i + \sum_i \hat{p}_i + \varepsilon}. \tag{2}$$

2.2 NOTEARS and IRM

NOTEARS fits a weighted adjacency W over $(P, v, t, h, \mathrm{VED}, \mathrm{defect_area}, \mathrm{regime})$:

$$\min_W \ \tfrac{1}{2n}\|X - XW\|_F^2 + \lambda_1 \|W\|_1 \quad \text{s.t. } \mathrm{tr}(e^{W \circ W}) - d = 0, \tag{3}$$

where the acyclicity constraint rules out directed cycles. After physics-consistency filtering (edges that violate the sign of $\partial\mathrm{VED}/\partial P$, etc., are removed), the recovered structure is checked against Eq. (1).

VED values split the dataset into two IRM environments: e_1 (stable, $30 \le \mathrm{VED} \le 80\,\mathrm{J/mm^3}$) and e_2 (low, $\mathrm{VED} < 30$). Training minimises

$$\min_\phi \sum_e \mathcal{L}^e(\phi) + \lambda_{\mathrm{IRM}} \left\| \nabla_{w|w=1} \mathcal{L}^e(\phi, w) \right\|^2, \tag{4}$$

with λ_{IRM} annealed from 0 to 10^3.

2.3 PINN and Neural ODE

The defect-nucleation ODE is taken as

$$\frac{df}{d\,\mathrm{VED}} = -\alpha(f - f_{\mathrm{eq}}(\mathrm{VED})), \tag{5}$$

where $f_{\mathrm{eq}} = f_{\max} \exp(-\beta(\mathrm{VED} - \mathrm{VED}_c)^2)$. A four-layer Tanh MLP maps VED to defect fraction and is trained with

$$\mathcal{L} = \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{phys}} \mathcal{L}_{\mathrm{ODE}}, \tag{6}$$

where $\mathcal{L}_{\mathrm{ODE}}$ is the mean-squared residual of Eq. (5) at collocation points. An ablation over $\lambda_{\mathrm{phys}} \in \{0, 0.1, 1, 10\}$ isolates the physics term.

The Neural ODE extends this to layer-wise dynamics,

$$\frac{df}{dl} = -\gamma\,\mathrm{VED}(l) \cdot f + s(\mathrm{VED}(l)), \tag{7}$$

comparing a pure-MLP vector field against one where the healing-rate structure is fixed by the network architecture.

3 Results

3.1 Hyperparameter Sweep

Table 1: Sweep results, four representative experiments.

Loss	lr	Dice	ROC AUC
BCEDice	5×10^{-4}	**0.787**	**0.9955**
Dice	1×10^{-4}	0.790	0.9941
BCE	1×10^{-4}	0.766	0.9938
Focal	1×10^{-4}	0.757	0.9921

Overlap-aware losses (BCEDice, Dice) outperform pixel-wise losses under the roughly 1% foreground fraction of the dataset. Pure Dice gives marginally higher Dice, but BCEDice is more reliable in practice because the BCE term penalises false positives on non-defective powder texture. At image level, the best configuration detects all 301 defective frames with zero misses and 4 false alarms. Threshold variation from $\tau = 0.3$ to 0.7 shifts Dice by less than 0.003, which confirms well-calibrated output probabilities.

3.2 NOTEARS and IRM

The recovered edges are: $t \to$ VED (+1.00), $h \to$ VED (+1.00), $v \to$ VED (-0.84), $P \to v$ (-0.91), VED $\to$ defect_area (+0.33). All signs match the analytic partial derivatives of Eq. (1). The $P \to v$ edge reflects the common operator practice of co-varying power and speed to hold a target VED.

IRM gives Dice = 0.633 on stable-VED images and 0.525 on low-VED images. The gap has a physical explanation: at stable VED the melt pool re-solidifies smoothly and defects appear as bounded dark regions, while at low VED the sintered powder retains inter-particle voids at the same spatial scale as LoF pores. The optimal threshold shifts from $\tau^* = 0.85$ (stable) to 0.90 (low VED), since the model assigns moderate confidence scores to ambiguous powder texture. This shift is not detectable without regime-stratified evaluation.

3.3 PINN

Table 2: PINN ablation: MSE against ODE ground truth.

λ_{phys}	MSE (vs. ODE)
0.0	highest
0.1	reduced
1.0	**lowest**
10.0	slightly higher

At $\lambda = 0$ the network fits the noisy training observations but oscillates outside the training range, violating the ODE. At $\lambda = 1$ the physics residual is small and the predicted nucleation curve matches the ODE solution across the full VED range, including the low-VED region where training data is sparse. At $\lambda = 10$ the physics constraint dominates and the network underfits near the LoF boundary. The physics-constrained Neural ODE converges faster than the pure-MLP version and generalises to initial conditions outside the training set, a direct result of the healing-rate structure being fixed by the architecture.

4 Discussion

The three mechanisms test different ways of using the same piece of physics. NOTEARS checks whether the governing equation is recoverable from data at all, and the answer here is yes, with edge signs consistent with Eq. (1). IRM uses the recovered structure to define training environments and produces regime-specific failure modes that trace back to process conditions. The PINN shows that even a one-dimensional physics constraint, evaluated at unlabelled collocation points, regularises the network toward the correct nucleation curve.

The IRM Dice scores are lower than the baseline. This is expected: the invariance penalty deliberately prevents the model from exploiting VED-regime-specific texture patterns, so raw Dice drops. The payoff is not metric gain but interpretability – the model fails in predictable, physically consistent ways across regimes.

Remaining limitations include the absence of high-VED keyhole data, the use of synthetic process-variable data in the NOTEARS stage (real per-image parameter measurements would strengthen the causal analysis), and the gap between the 1-D PINN model and actual image-level defect prediction. Closing these gaps is left for future work.

References

[1] T. DebRoy et al., "Additive manufacturing of metallic components," *Prog. Mater. Sci.*, vol. 92, pp. 112–224, 2018.

[2] O. Ronneberger, P. Fischer, T. Brox, "U-Net," *MICCAI*, pp. 234–241, 2015.

[3] X. Zheng et al., "DAGs with NO TEARS," *NeurIPS*, pp. 9472–9483, 2018.

[4] M. Arjovsky et al., "Invariant risk minimization," *arXiv:1907.02893*, 2019.

[5] T.-Y. Lin et al., "Focal loss for dense object detection," *ICCV*, pp. 2980–2988, 2017.

[6] M. Raissi, P. Perdikaris, G. Karniadakis, "Physics-informed neural networks," *J. Comput. Phys.*, vol. 378, pp. 686–707, 2019.

[7] R. T. Q. Chen et al., "Neural ordinary differential equations," *NeurIPS*, 2018.